

Supermarket Sales Dataset

Pandas + Matplotlib + Seaborn Practice Operations

Run these operations in Jupyter Notebook / Google Colab / VS Code.

1. Read CSV

```
df = pd.read_csv("supermarket_sales.csv")  
df
```

2. First 5 Rows

```
df.head()
```

3. Last 5 Rows

```
df.tail()
```

4. Shape

```
df.shape
```

5. Column Names

```
df.columns
```

6. Data Types

```
df.dtypes
```

7. Missing Values

```
df.isnull().sum()
```

8. Statistics

```
df.describe()
```

9. Total Sales

```
df["Total_Sales"].sum()
```

10. Average Sales

```
df["Total_Sales"].mean()
```

11. Highest Sale

```
df["Total_Sales"].max()
```

12. Lowest Sale

```
df["Total_Sales"].min()
```

13. Highest-Selling Order

```
df.loc[df["Total_Sales"].idxmax()]
```

14. Sales Above 10000

```
df[df["Total_Sales"] > 10000]
```

15. Pune Orders

```
df[df["City"] == "Pune"]
```

16. Electronics Orders

```
df[df["Category"] == "Electronics"]
```

17. Sort by Sales

```
df.sort_values("Total_Sales", ascending=False)
```

18. Total Sales by City

```
df.groupby("City")["Total_Sales"].sum()
```

19. Average Sales by Category

```
df.groupby("Category")["Total_Sales"].mean()
```

20. Quantity by Product

```
df.groupby("Product")["Quantity"].sum()
```

21. Sales by Payment Method

```
df.groupby("Payment_Method")["Total_Sales"].sum()
```

22. Matplotlib Bar Chart

```
city_sales=df.groupby("City")["Total_Sales"].sum()  
plt.bar(city_sales.index, city_sales.values)  
plt.xlabel("City"); plt.ylabel("Total Sales"); plt.title("Sales by City"); plt.show()
```

23. Matplotlib Line Chart

```
plt.plot(df["Date"], df["Total_Sales"], marker="o")  
plt.xticks(rotation=90); plt.xlabel("Date"); plt.ylabel("Total Sales"); plt.title("Sales Trend"); plt.show()
```

24. Matplotlib Scatter Plot

```
plt.scatter(df["Quantity"], df["Total_Sales"])  
plt.xlabel("Quantity"); plt.ylabel("Total Sales"); plt.title("Quantity vs Sales"); plt.show()
```

25. Matplotlib Histogram

```
plt.hist(df["Total_Sales"], bins=6)  
plt.xlabel("Total Sales"); plt.ylabel("Orders"); plt.title("Sales Distribution"); plt.show()
```

26. Matplotlib Pie Chart

```
payment=df["Payment_Method"].value_counts()  
plt.pie(payment, labels=payment.index, autopct="%1.1f%%")  
plt.title("Payment Method Distribution"); plt.show()
```

27. Seaborn Count Plot

```
sns.countplot(data=df, x="Category")  
plt.title("Orders by Category"); plt.xticks(rotation=30); plt.show()
```

28. Seaborn Bar Plot

```
sns.barplot(data=df, x="City", y="Total_Sales", estimator=sum)  
plt.title("Total Sales by City"); plt.show()
```

29. Seaborn Box Plot

```
sns.boxplot(data=df, x="Category", y="Total_Sales")  
plt.title("Sales Distribution by Category"); plt.xticks(rotation=30); plt.show()
```

30. Seaborn Scatter Plot

```
sns.scatterplot(data=df, x="Quantity", y="Total_Sales", hue="Category")  
plt.title("Quantity vs Sales"); plt.show()
```

31. Seaborn Heatmap

```
corr=df.select_dtypes("number").corr()  
sns.heatmap(corr, annot=True)  
plt.title("Correlation Heatmap"); plt.show()
```

32. Seaborn Pair Plot

```
sns.pairplot(df[["Quantity", "Unit_Price", "Total_Sales"]])  
plt.show()
```

33. Create New Column

```
df["Calculated_Sales"]=df["Quantity"]*df["Unit_Price"]  
df.head()
```

34. Top 5 Orders

```
df.nlargest(5, "Total_Sales")
```

35. Save Result

```
df.to_csv("sales_analysis_result.csv", index=False)
```

Practice Questions

1. Which city has the highest total sales?
2. Which product generated the highest sales?
3. Which category has the most orders?
4. Which payment method is most common?
5. Does quantity have a relationship with total sales?
6. Which category has the highest average order value?
7. Which are the top 5 orders by sales?
8. What does the correlation heatmap tell you?